MongoDB Learning Path

1. MongoDB Basics

Learn what it is, not how to code yet.

What is MongoDB

- NoSQL database
- Document-oriented database
- BSON
- Schema vs schema-less
- When to use MongoDB

2. Core Terminology (Very Important)

Write definitions for each.

- **Database**
- **Collection**
- **Document**
- **Field**
- **_id**
- **ObjectId**
- **Namespace**

3. Data Model Concepts

Only concepts, no queries.

- Embedded documents
- Referenced documents
- One-to-one relationship
- One-to-many relationship
- Many-to-many relationship

- Denormalization
- Normalization (MongoDB context)

4. Data Types in MongoDB

Definitions only.

- String
- Number
- Boolean
- Array
- Object
- Date
- ObjectId
- Null

5. CRUD Concept Overview

High-level meaning only.

- Create
- Read
- Update
- Delete
- Insert vs update
- Update vs replace

6. Query Fundamentals (Conceptual)

No syntax yet.

- Filter
- Projection
- Sorting
- Limiting
- Skipping
- Cursor

7. Indexes (Very High Value)

Just concepts.

- What is an index
- Why indexes are needed
- Single field index
- Compound index
- Index cost (trade-off)

8. Schema Design Principles

MongoDB-specific thinking.

- Data access patterns
- Read-heavy vs write-heavy
- Embedding vs referencing
- Avoiding over-normalization

9. Consistency & Reliability (Light)

Basic awareness.

- Atomic operations
- Document-level atomicity
- Transactions (concept only)
- Write concern (definition)

10. MongoDB in Web Apps (Context Only)

No code.

- MongoDB in backend
- Why not store external API data
- User-owned data concept
- ObjectId vs external IDs